Calculus: Basic Concepts for High School (L.V. Tarasov)

Dialogue One: Infinite Numerical Sequence

Author: Let us start our discussions of calculus by considering the definition of an infinite numerical sequence or simply a sequence.

We shall consider the following examples of sequences:

$1, 2, 4, 8, 16, 32, 64, 128, \ldots\ldots$ (1)

$5, 7, 9, 11, 13, 15, 17, 19, \ldots\ldots$ (2)

$1, 4, 9, 16, 25, 36, 49, 64, \ldots\ldots$ (3)

$1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \sqrt{6}, \sqrt{7}, 2\sqrt{2}, \ldots\ldots$ (4)

$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{8}{9}, \ldots\ldots$ (5)

$2, 0, -2, -4, -6, -8, -10, -12, \ldots\ldots$ (6)

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, \ldots\ldots$ (7)

$1, \frac{1}{2}, 3, \frac{1}{4}, 5, \frac{1}{6}, 7, \frac{1}{8}, \ldots\ldots$ (8)

$1, -1, \frac{1}{3}, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{5}, \frac{1}{7}, -\frac{1}{7}, \ldots\ldots$ (9)

$1, \frac{2}{3}, \frac{1}{3}, \frac{4}{4}, \frac{1}{5}, \frac{6}{7}, \frac{1}{7}, \frac{8}{9}, \ldots\ldots$ (10)

Have a closer look at these examples. What do they have in common?

Sam: • All of these sequences go on forever, i.e. they are infinite.
• We can also predict the $(n+1)$th term for each sequence given the first n terms, with the help of an implicit law.

Reader: It is assumed that in each example there must be an infinite number of terms in a sequence. But in general, they are all different.

Author: In each example we have eight terms of a sequence. Could you write, say, the ninth term?

~~Reader:~~ **Sam:** Seq 1 $\rightarrow 256$ Seq 2 $\rightarrow 21$ Seq 3 $\rightarrow 81$ Seq 4 $\rightarrow \sqrt{9} = 3$

Seq 5 $\rightarrow \frac{9}{10}$ Seq 6 $\rightarrow -14$ Seq 7 $\rightarrow \frac{1}{9}$ Seq 8 $\rightarrow 9$ Seq 9 $\rightarrow \frac{1}{9}$

Seq 10 $\rightarrow \frac{1}{9}$